

Homework - Neural networks - Part A (35 points)

Interactive activation and competition

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Note:

Part A is short answer without any code. Please complete the responses to these questions as a markdown cell inserted beneath the question prompts.

In this assignment, you will get hands on experience with a classic neural network model of memory known as the Interactive Activation and Competition (IAC) model. We will go through a series of exercises that will stretch your understanding of the IAC model in various ways. The exercises below examine how the mechanisms of interactive activation and competition can be used to illustrate two key properties of human memory:

- Retrieval by name and by content.
- Spontaneous generalization over a set of familiar items.

These exercises are from Chapter 2 of the [online PDP Handbook](#) by James McClelland.

You should review the slides from lecture and read Section 2.1 of the PDP Handbook before continuing. This has important background and technical details on how the IAC model works. The IAC model instantiates knowledge that someone may have from watching the 1960s musical "West Side Story," where two gangs the "Jets" and "Sharks" struggle for neighborhood control in Manhattan. The "database" for this exercise is the Jets and Sharks data base shown in Figure 1, which has the central characters from the two gangs. You are to use the IAC model in conjunction with this data base to run illustrative simulations of these basic properties of memory.

The Jets and The Sharks

Name	Gang	Age	Edu	Mar	Occupation
Art	Jets	40's	J.H.	Sing.	Pusher
Al	Jets	30's	J.H.	Mar.	Burglar
Sam	Jets	20's	COL.	Sing.	Bookie
Clyde	Jets	40's	J.H.	Sing.	Bookie
Mike	Jets	30's	J.H.	Sing.	Bookie
Jim	Jets	20's	J.H.	Div.	Burglar
Greg	Jets	20's	H.S.	Mar.	Pusher
John	Jets	20's	J.H.	Mar.	Burglar
Doug	Jets	30's	H.S.	Sing.	Bookie
Lance	Jets	20's	J.H.	Mar.	Burglar
George	Jets	20's	J.H.	Div.	Burglar
Pete	Jets	20's	H.S.	Sing.	Bookie
Fred	Jets	20's	H.S.	Sing.	Pusher
Gene	Jets	20's	COL.	Sing.	Pusher
Ralph	Jets	30's	J.H.	Sing.	Pusher
Phil	Sharks	30's	COL.	Mar.	Pusher
Ike	Sharks	30's	J.H.	Sing.	Bookie
Nick	Sharks	30's	H.S.	Sing.	Pusher
Don	Sharks	30's	COL.	Mar.	Burglar
Ned	Sharks	30's	COL.	Mar.	Bookie
Karl	Sharks	40's	H.S.	Mar.	Bookie
Ken	Sharks	20's	H.S.	Sing.	Burglar
Earl	Sharks	40's	H.S.	Mar.	Burglar
Rick	Sharks	30's	H.S.	Div.	Burglar
Ol	Sharks	30's	COL.	Mar.	Pusher
Neal	Sharks	30's	H.S.	Sing.	Bookie
Dave	Sharks	30's	H.S.	Div.	Pusher

Figure 1: Characteristics of a number of individuals belonging to two gangs, the Jets and the Sharks. (From "Retrieving General and Specific Knowledge From Stored Knowledge of Specifics" by J. L. McClelland, 1981, Proceedings of the Third Annual Conference of the Cognitive Science Society.)

Software and architecture

We will be using IAC software from Todd Gureckis which you access [here](#).

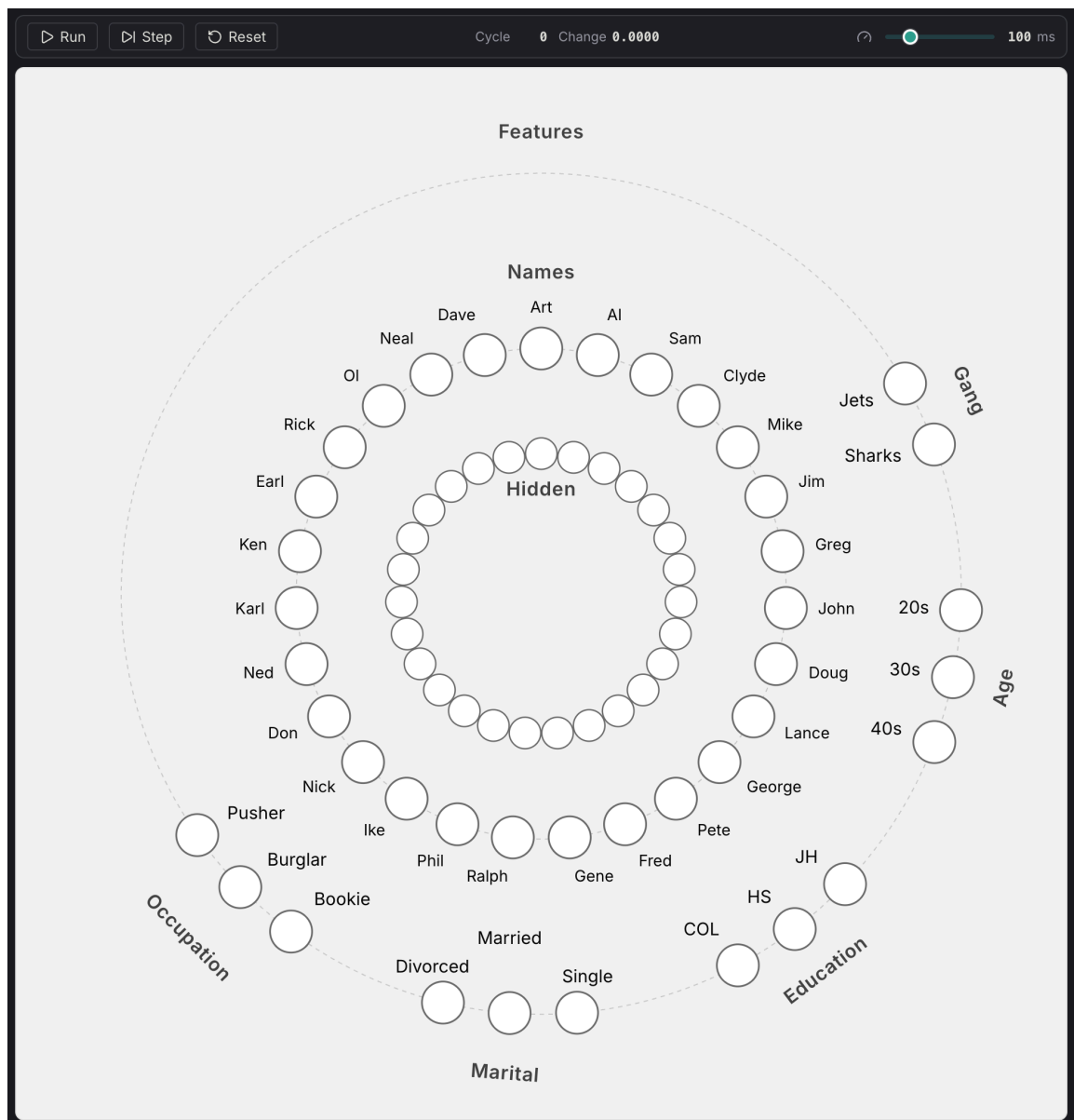


Figure 2: Screen shot from Gureckis's software. Units are organized into 7 groups.

You will see a display that looks like Figure 2. The units are grouped into seven pools: a pool of *visible* name units, a pool of *gang* units, a pool of *age* units, a pool of *education* units, a pool of *marital status* units, a pool of *occupation* units, and a pool of *hidden* units. The name pool contains a unit for the name of each person; the gang pool contains a unit for each of the gangs the people are members of (Jets and Sharks); the age pool contains a unit for each age range; and so on. Finally, the *hidden* pool contains an instance unit for each individual in the set.

The units in the first six pools can be called visible units, since all are assumed to be accessible from outside the network. Those in the gang, age, education, marital status, and occupation pools can also be called property units. The instance units are assumed to be inaccessible, so they can be called hidden units.

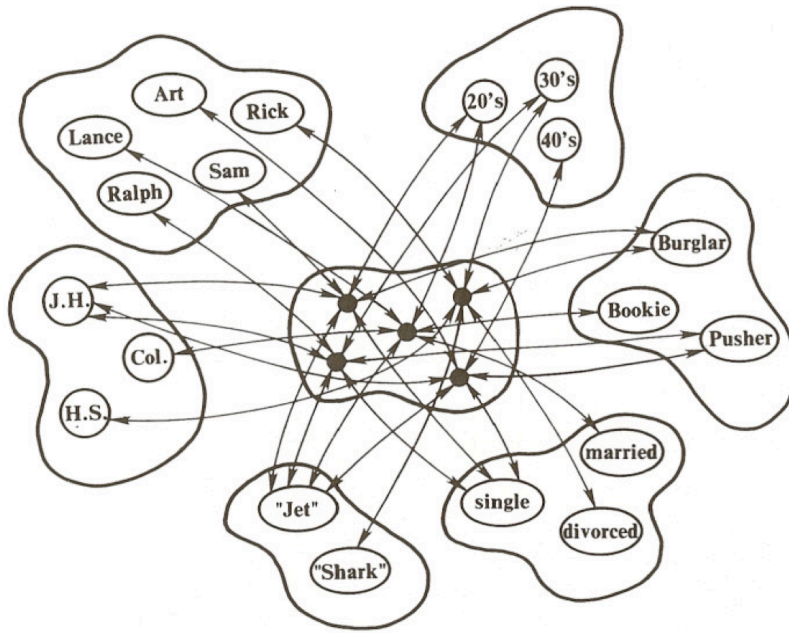


Figure 3: The units and connections for some of the individuals in Figure 1. The arrows represent excitatory connections. The outlined groups of units have mutually inhibitory connections (not shown). (From "Retrieving General and Specific Knowledge From Stored Knowledge of Specifics" by J. L. McClelland, 1981, *Proceedings of the Third Annual Conference of the Cognitive Science Society*.)

Each unit has an inhibitory connection to every other unit in the same pool. In addition, there are two-way excitatory connections between each instance unit and the units for its properties, as illustrated in Figure 3. Note that the figure is incomplete, in that only some of the name and instance units are shown. These names are given only for the convenience of the user, of course; all actual computation in the network occurs only by way of the connections. You can also view the different connections using the IAC software by hovering your mouse over a particular unit (Figure 4).

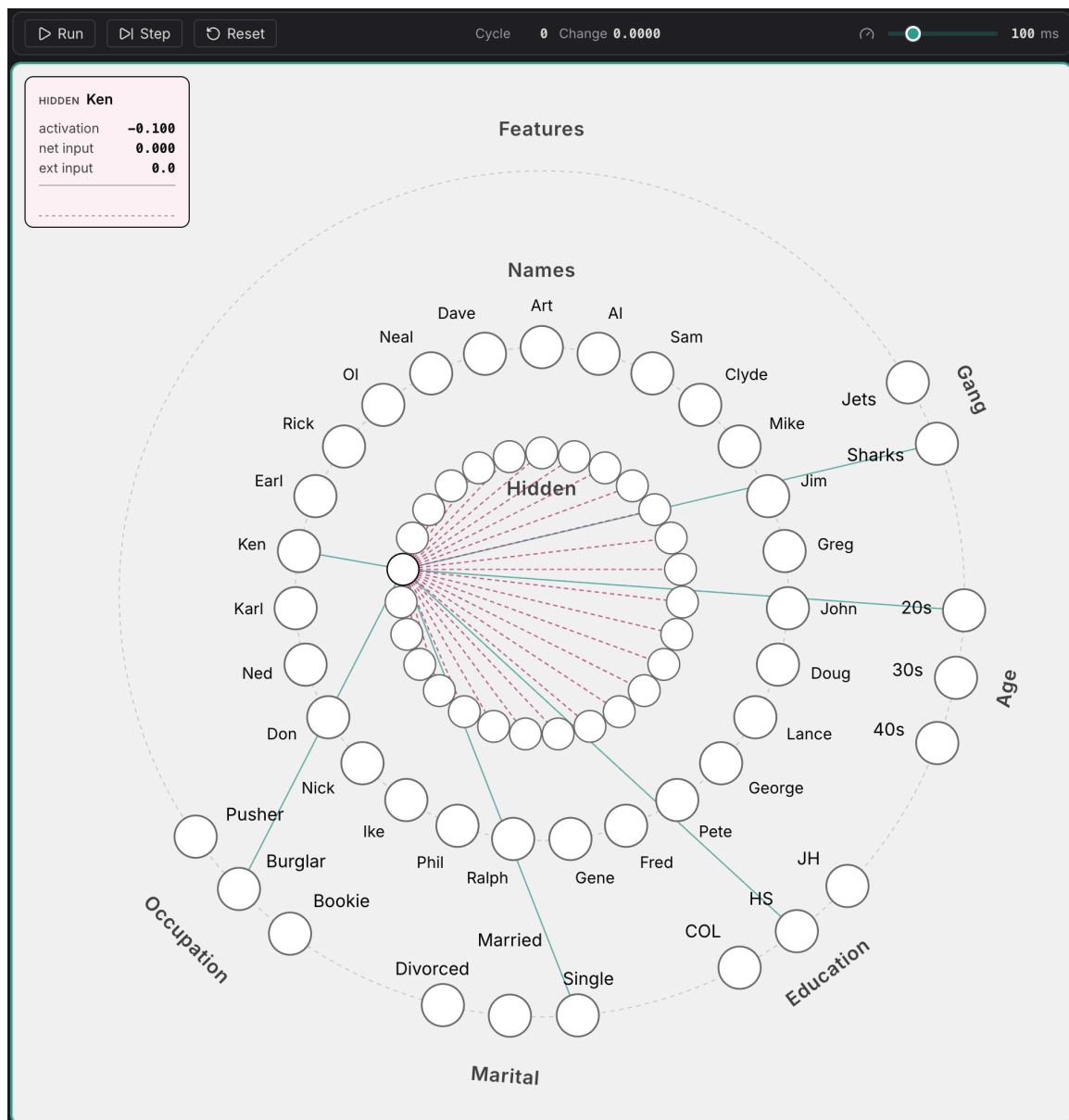


Figure 4: You can view the connections to a unit by placing your mouse over it. Green connections are excitatory and red dotted connections are inhibitory.

Since everything is set up for you, you are now ready to do each of the separate parts of the exercise. Each part is accomplished by using the interactive activation and competition process to do pattern completion, given some probe that is presented to the network. For example, to retrieve an individual's properties from his name, you simply provide external input to his name unit, then allow the IAC network to propagate activation first to the name unit, then from there to the instance units, and from there to the units for the properties of the instance.

Exercise: Retrieving an individual from his name

To illustrate retrieval of the properties of an individual from his name, we will use Ken as our example. Make sure the simulation is paused (press SPACE) and press 'r' to reset it.

(Or click from the menu on top.) Set the external input of Ken's name unit to 1 by clicking on the name unit (not the hidden unit!). The circle should be highlighted in yellow to indicate the external input.

A unit's activation level can be visualized by the color. The baseline activation of -0.1 is shown as white. Higher activation than baseline is shown as pinkish, and lower activation than baseline is shown as blueish. A unit's precise activation can be examined by rolling the mouse over the unit.

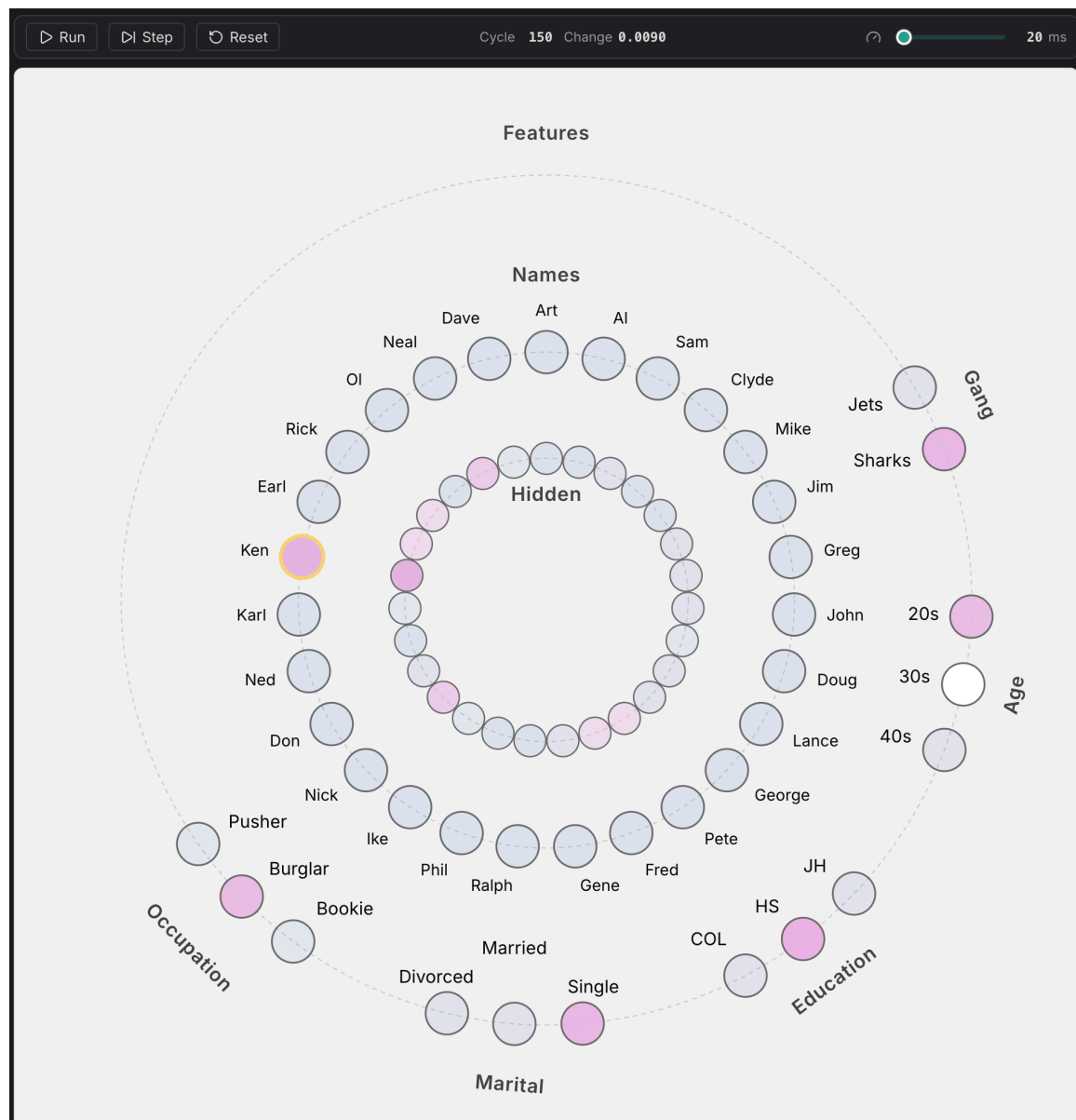


Figure 5: The display screen after about 150 cycles with external input to the name unit for Ken.

Press SPACE to unpause and allow the network to run for approximately 150 cycles (the cycle counter is at the top of the panel). Be sure to pause at about 150 cycles! You can also click Step for more granularity. A picture of the screen after 150 cycles is shown in Figure 5. At this point, you can check to see that the model has indeed retrieved the

pattern for Ken correctly. There are also several other things going on that are worth understanding. Answer all of the following questions below regarding the network at this state (you'll have to refer to the properties of the individuals, as given in Figure 1).

Problem 1 (10 points)

None of the visible name units other than Ken were activated, yet a few other hidden instance units are active (i.e., their activation is greater than 0). Explain why these units are active. Keep your response short (about 3 sentences).

Answer: Since Ken's instance units activate his property units ie the attributes that are fully lit up, those units send excitation back to every instance unit connected to them, not just Ken's. The people who share the most properties with Ken get enough of this feedback which is shown as the levels rise to above 0. This is shown as Nick, Neal, Rick, Pete and Fred have all levels above 0. This makes sense since Nick and Neal share both Sharks, HS, and Single, whereas Rick is a Shark, HS, and Burglur. No other name units turn on because Ken's instance unit inhibits the others, so they stay only partly active and send little activation out to their names.

Problem 2 (10 points)

Some of Ken's properties are activated more strongly than others. Why? Keep your response short (about 3 sentences).

Each property unit gets inputs from Ken's instance plus any any other active instances connected to it, ie those it has in common with others. Sharks and HS are shared by all three similar people (Nick, Neal, Rick), so they get most support. Single (Nick and Neal) and Burglur (Rick) get less. 20s is the weakest since none of the other active people are in their 20s. All three are in their 30s, so the 30s unit rises.

Retrieval from a partial description

Next, we will use the IAC software to illustrate how it can retrieve an instance from a partial description of its properties. We will continue to use Ken, who, as it happens, can be uniquely described by two properties, Shark and in 20s. Reset the network ('r') and make sure everything is paused and that all units have input of 0. Click to set the external input of the Sharks unit and the in 20s unit to 1.00. Run a total of 150 cycles again, and take a look at the state of the network.

Of all of the visible name units, Ken's name should be the most active. Compare this state of the network with the screen shot of the previous state of the network when activating Ken's name directly, such as that in Figure 5.

Problem 3 (10 points)

Explain why the occupation units show partial activations of units other than Ken's occupation, which is Burglar. While being succinct, try to get to the bottom of this, and contrast the current case with the previous case. Keep your response short (about 3 sentences).

Without the name unit, Ken's instance has no unique input on its own. It is just the one person who matches both probes, so instances matching one probe plus the properties Ken turns on (HS, Single) stay fairly active. These are Nick and Neal (Sharks) and Pete and Fred (20s). Those four are Pushers (Nick, Fred) and Bookies (Neal, Pete), and none are Burglurs. So Pusher and Bookie each get input from two instances, which nearly matches Burglur's input from Ken alone. When the name was the input, Ken's instance dominated more strongly. The similar people then split across occupations, so Burglur easily beat the others and pushed them below baseline.

Spontaneous generalization

Now we consider the network's ability to retrieve appropriate generalizations over sets of individuals—that is, its ability to answer questions like “What are Jets like?” or “What are people who are in their 20s and have only a junior high education like?” Reset ('r') the network. Set the external input of Jets to 1.00 by clicking on it. Run the network for 150 cycles and observe what happens.

Problem 4 (5 points)

Given the network's state, what can you infer about a typical Jet? (1-2 sentences is plenty).

A typical Jet is in his 20s, has a junior high education, and is single. Those units become strongly active even though no single Jet was probed, because most Jets have them (9 of 15 for each). The occupation units stay weak and about equal, because the Jets are split evenly: 5 Pushers, 5 Burglars and 5 Bookies. So the network doesn't predict a typical job.